

Measurement

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Measurement: The Foundation of Scientific Research



Methods in Psychological Research

Roadmap

- + Measurement
- + Levels of measurement
- + Goals of measurement



Understanding Measurement

- + Definition: Measurement is the assignment of numbers to characteristics of people or objects
 - + 4th step in the research process
 - + It's purpose is to describe and differentiate
- + Examples: measurement scales
 - + Speed → miles-per-hour
 - + Temperature → Kelvin (K)
 - + Order of finishers in a race → 1st, 2nd, 3rd, etc.
 - + Numbers on the back of basketball jerseys → 28, 156, 152, 156, 215



Variables: The Building Blocks of Measurement

- + Variables contain the outcome of measurement processes
- + Variables can be:
 - + Qualitative:
 - + Numbers represent qualities (not quantities)
 - + Example: Coding for eye color (1 = Blue, 2 = Brown, 3 = Green)
 - + Quantitative:
 - + Numbers mean something in relation quantities in the real world
 - + Example: Height in centimeters, Weight in kilograms



Levels of Measurement



Stevens' Classification (1946):

+ Characterizes each level of measurement

- + Nominal
- + ordinal,
- + interval,
- + ratio

+ Key Properties:

- + Identity
- + Order
- + Equal Intervals
- + Absolute Zero



Stevens, Stanley Smith (1946). On the theory of scales of measurement. *Science*, 103(2684), 677-680.

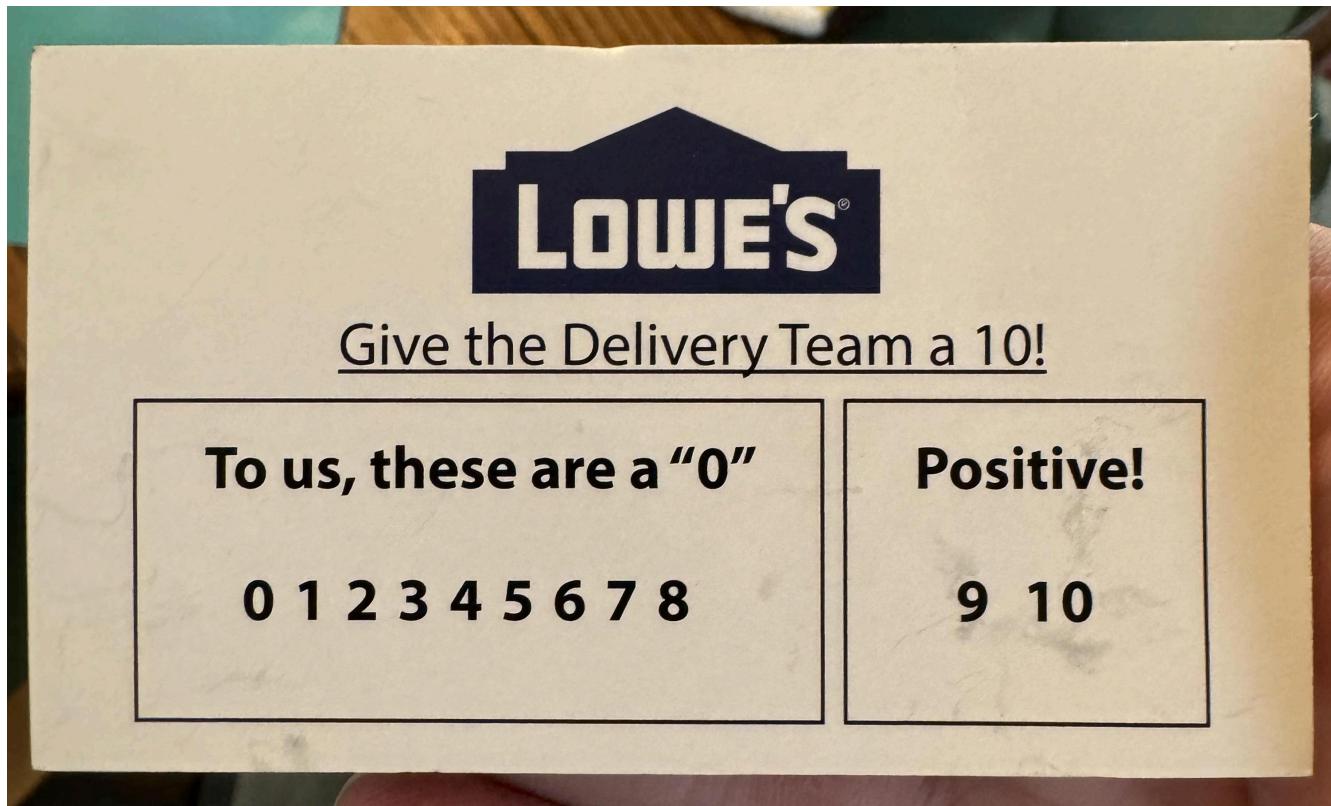


Key Properties of Measurement

- + Have absolute zero
 - + 0 indicated absence (origin means zero)
- + Equal intervals
 - + An interval means the same value at any point on measurement scales
- + Order
 - + Number means order
- + Identity
 - + Different numbers mean different measurement outcomes ($1 \neq 2$)



Which properties does this measure have?



Source: [/u/ZigeonV0](#) via [r/assholedesign](#)



Methods in Psychological Research

Ratio (quantitative)

- + All four properties
 - + Have absolute zero
 - + Equal intervals
 - + Order
 - + Identity
- + Multiplication and division are permissible transformations
- + Examples:
 - + Height, weight, age, income, time, etc.



Example (Base R)

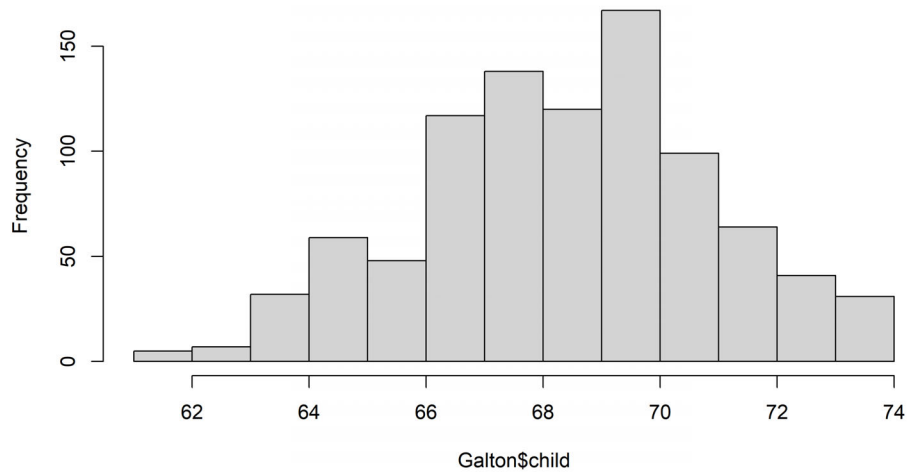
```
# loads the HistData package
library(HistData)
# loads the Galton dataset
data("Galton")
# First 3n rows of data
head(Galton, n=3)
```

```
##      parent child
## 1      70.5  61.7
## 2      68.5  61.7
## 3      65.5  61.7
```



Example (Base R)

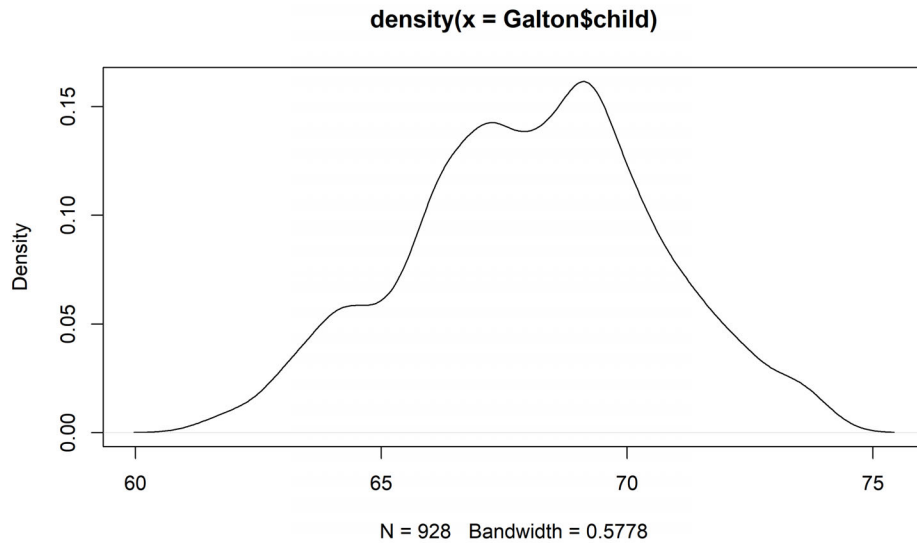
Histogram of Galton\$child



```
# Histogram  
hist(Galton$child)
```



Example (Base R)



```
# Density Plot  
plot(density(Galton$child))
```



Bandwidth Aside

Bandwidth: Smoothing Method

```
args(density.default)
```

```
## function (x, bw = "nrd0", adjust = 1, kernel = c("gaussian",  
##         "epanechnikov", "rectangular", "triangular", "biweight",  
##         "cosine", "optcosine"), weights = NULL, window = kernel,  
##         width, give.Rkern = FALSE, subdensity = FALSE, warnWbw = var(weights)  
##         0, n = 512, from, to, cut = 3, na.rm = FALSE, ...)  
## NULL
```

```
#?density # Gives you documentation
```



Bandwidth Aside

Usage

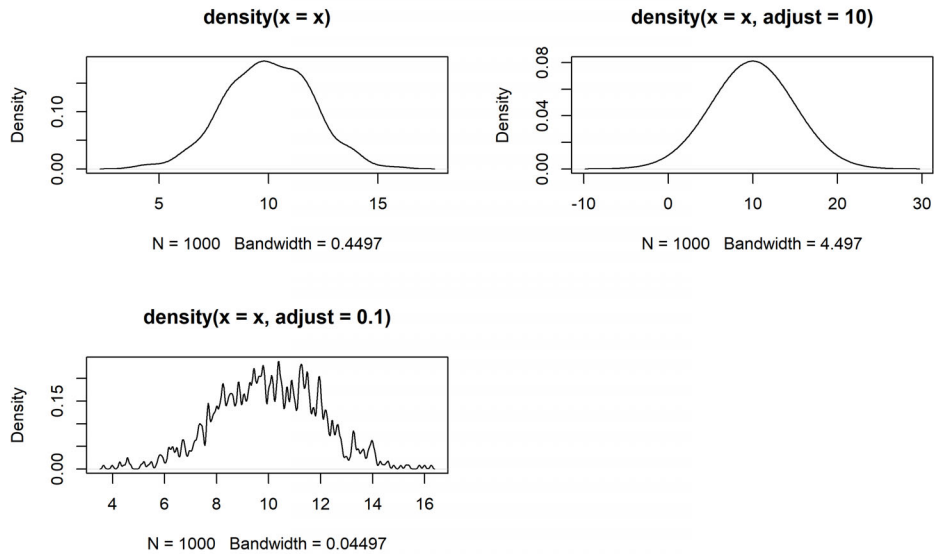
```
density(x, ...)
## Default S3 method:
density(x, bw = "nrd0", adjust = 1,
        kernel = c("gaussian", "epanechnikov", "rectangular",
                    "triangular", "biweight",
                    "cosine", "optcosine"),
        weights = NULL, window = kernel, width,
        give.Rkern = FALSE,
        n = 512, from, to, cut = 3, na.rm = FALSE, ...)
```

Arguments

x	the data from which the estimate is to be computed. For the default method a numeric vector: long vectors are not supported.
bw	the smoothing bandwidth to be used. The kernels are scaled such that this is the standard deviation of the smoothing kernel. (Note this differs from the reference books cited below, and from S-PLUS.) bw can also be a character string giving a rule to choose the bandwidth. See bw.nrd . The default, "nrd0", has remained the default for historical and compatibility reasons, rather than as a general recommendation, where e.g., "SJ" would rather fit, see also Venables and Ripley (2002). The specified (or computed) value of bw is multiplied by adjust.
adjust	the bandwidth used is actually adjust*bw. This makes it easy to specify values like 'half the default' bandwidth.
kernel, window	a character string giving the smoothing kernel to be used. This must partially match one of "gaussian", "rectangular", "triangular", "epanechnikov", "biweight", "cosine" or "optcosine", with default "gaussian", and may be abbreviated to a unique prefix (single letter). "cosine" is smoother than "optcosine", which is the usual 'cosine' kernel in the literature and almost MSE-efficient. However, "cosine" is the version used by S.
weights	numeric vector of non-negative observation weights, hence of same length as x. The default NULL is equivalent to <code>weights = rep(1/nx, nx)</code> where nx is the length of (the finite entries of) x[].



Bandwidth Aside



```
set.seed(201010)
x <- rnorm(1000, 10, 2)
par(mfrow = c(2,2))

#A bit bumpy
plot(density(x))
#Very sooth
plot(density(x,adjust = 10))
#Very bumpy
plot(density(x,adjust = .1))
```



Interval (quantitative)

- + Has order, identity, and equal intervals
 - + (all but absolute zero)
- + Addition is a permissible transformation
- + Example: Temperature in Celsius or Fahrenheit

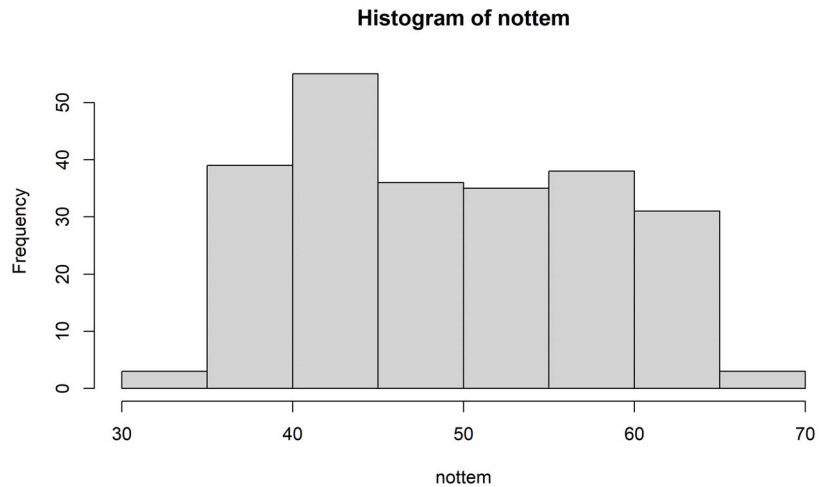
```
# Interval Example
library(datasets)
data("nottem")
nottem[1:10]                                # First ten rows of data
```

```
##      [1] 40.6 40.8 44.4 46.7 54.1 58.5 57.7 56.4 54.3 50.5
```

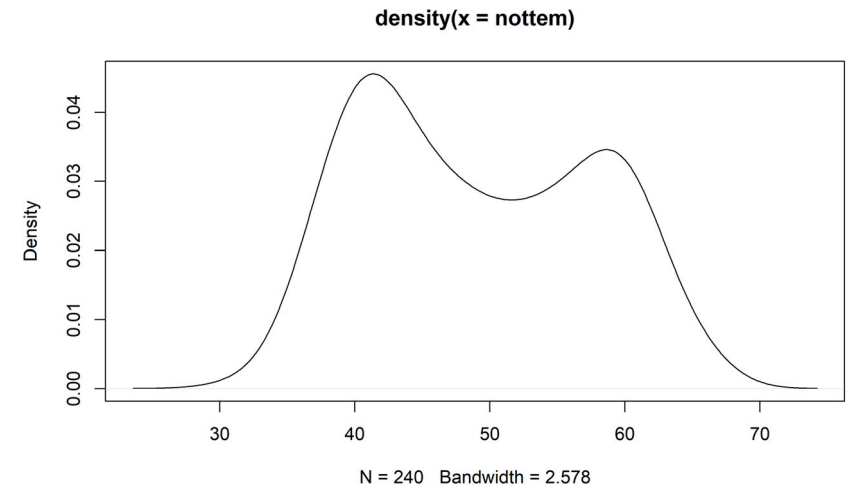


Interval (quantitative)

```
# Histogram  
hist(nottem)
```



```
# Density Plot  
plot(density(nottem))
```



Ordinal (qualitative)

- + Has order and identity
- + Monotonic transformations are permissible
- + These variables maintain the order of the values

```
# Ordinal Example
library(ggplot2movies)
data(movies)
# First 4 rows of data, with a non-missing budget
head(movies[!is.na(movies$budget),], n = 14)
```

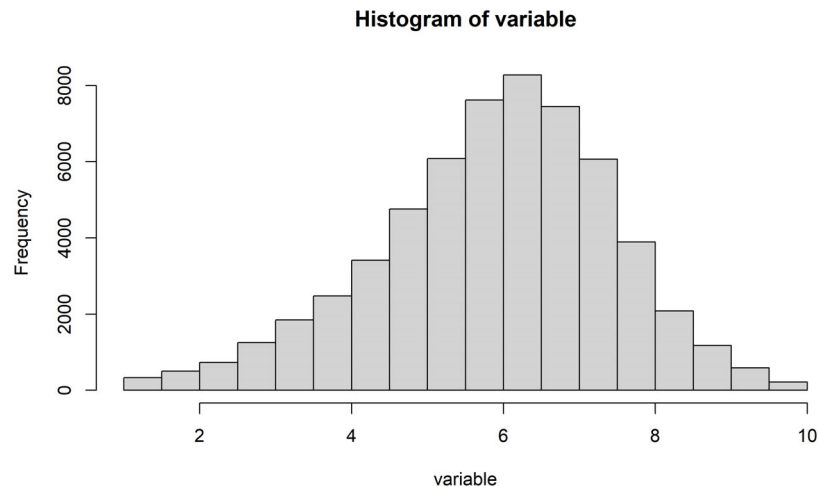
- + Example: Education levels (High School, Bachelor's, Master's, PhD)

```
## # A tibble: 14 × 24
##   title    year length budget rating votes    r1
##   <chr>  <int>  <int>  <int>  <dbl> <int> <dbl> <dbl>
## 1 'G' M...  1935    85 4.50e5    7.2   281    0    4
## 2 'Mano...  1966   74 1.9 e4    1.6  7996   74.5   4.5   4.5   4.5
## 3 'Til ...  1997  113 2.3 e7    4.8   799    4.5   4.5   4.5  14.5
## 4 .com ...  2002   96 5 e6     3.7   271   64.5   4.5   4.5   4.5
## 5 10 Th...  1999   97 1.6 e7    6.7 19095    4.5   4.5   4.5   4.5
## 6 100 M...  2002   98 1.10e6    5.6   181    4.5   4.5   4.5   4.5
## # i 8 more rows
## # Methods in Psychological Research
## "" ""
```

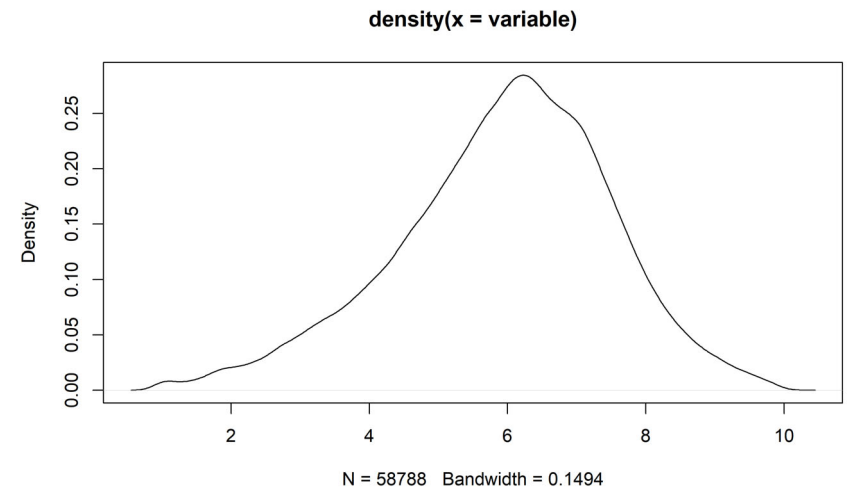


Ordinal (qualitative)

```
# Histogram  
variable <- movies$rating  
hist(variable)
```



```
# Density Plot  
plot(density(variable))
```



Nominal

- + Only has identity
- + Any identity-preserving transformation is permissible
- + Example: Jersey numbers in sports

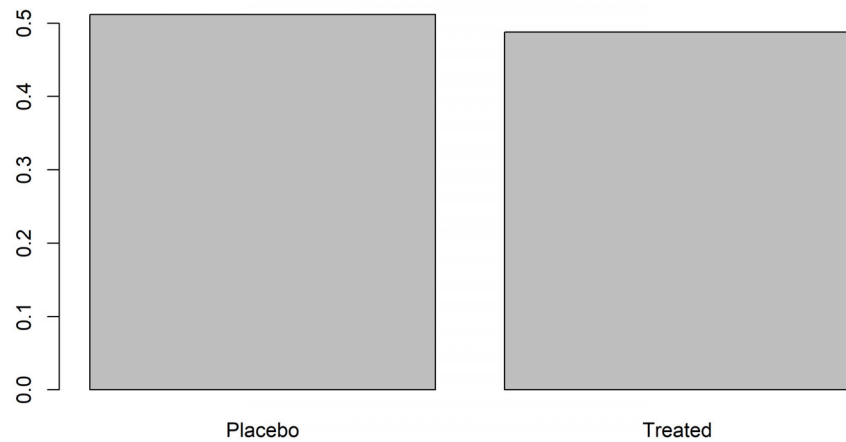
```
# Nominal Example
library(vcd)
data(Arthritis)
head(Arthritis[, c("ID", "Treatment")], 8)
```

```
##      ID Treatment
## 1  57    Treated
## 2  46    Treated
```

```
## 3  77    Treated
## 4  17    Treated
## 5  36    Treated
## 6  23    Treated
## 7  75    Treated
## 8  39    Treated
```



Basic Barplot



```
variable<-Arthritis$Treatment  
#hist(variable) # error  
  
barplot_fix <-  
  prop.table(table(variable))  
# Sometimes, R is silly  
barplot(barplot_fix)
```



More complex measurement level taxonomy

- + Missing (considered nominal under the Stevens)
- + Binary (considered nominal under the Stevens)
- + Nominal (considered nominal under the Stevens)
- + Partially ordered (considered ordinal under Stevens)
- + Fully ordered (considered ordinal under Stevens)
- + Interval
- + Ratio
- + Absolute measurement (has no permissible transformation)
 - + 6.02×10^{23}
 - + π



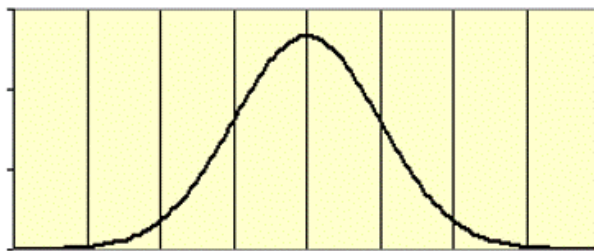
Could measurement level be itself on a continuum?

+ Example: IQ

- + Falls between interval and ratio?
- + Or falls between ordinal and interval?

IQ Comparison Site
www.iqcomparisonsite.com
Copyright 2007 Rodrigo de la Jara

IQ Normal Curve



Standard Deviations	-4	-3	-2	-1	0	1	2	3	4
Wechsler IQ	40	55	70	85	100	115	130	145	160
Stanford-Binet IQ	36	52	68	84	100	116	132	148	164
Cumulative %	0.003	0.135	2.275	15.866	50.000	84.134	97.725	99.865	99.997

1 Std - Medicals+ ·

85 IQ (top 86%) 😊 who can top that? 😊

CERTIFICATE OF COMPLETION
IS AWARDED TO

PERCENTILE
14.31%

DATE
01/27/2016

My IQ
85

Caitlin

▶ **START THE TEST** ◀

The average IQ in United Kingdom is 100. And you?
Take the test now!

45 questions, 15 minutes, automatically scored at the end of the test. The next session begins now, click here to start. 10 minutes left.

IQ-RESEARCH.ORG

👍 Gefällt mir 💬 Kommentieren ➦ Teilen

Sara Uhhh I don't think that means what you think it means... 😊

Gefällt mir · Antworten · Gerade eben



Goals of Measurement



Goals of Measurement

+ Reliability

- + "...the degree to which a test or measure produces the same scores when applied in the same circumstances..." (Thomas and Nelson 1996)
- + In other words, if you take the measure again, will you get the same result?

+ (Internal) Validity

- + "Degree to which a test or instrument measures what it purports to measure" (Thomas and Nelson 1996)
- + In other words, does your measure measure what is it supposed to measure?



Types of Validity (More on this later...)

- + Two Major Areas within Validity
 - + Internal Validity
 - + Is this evidence supportive of our claim, within this study?
 - + External Validity
 - + Is this evidence supportive of our claim beyond this study?
 - + Does this finding generalize to outside this study?



Practical Applications

- + Psychology: Measuring personality traits
- + Education: Assessing student performance
- + Medicine: Tracking patient vital signs
- + Business: Evaluating customer satisfaction
- + Sports: Analyzing athletic performance



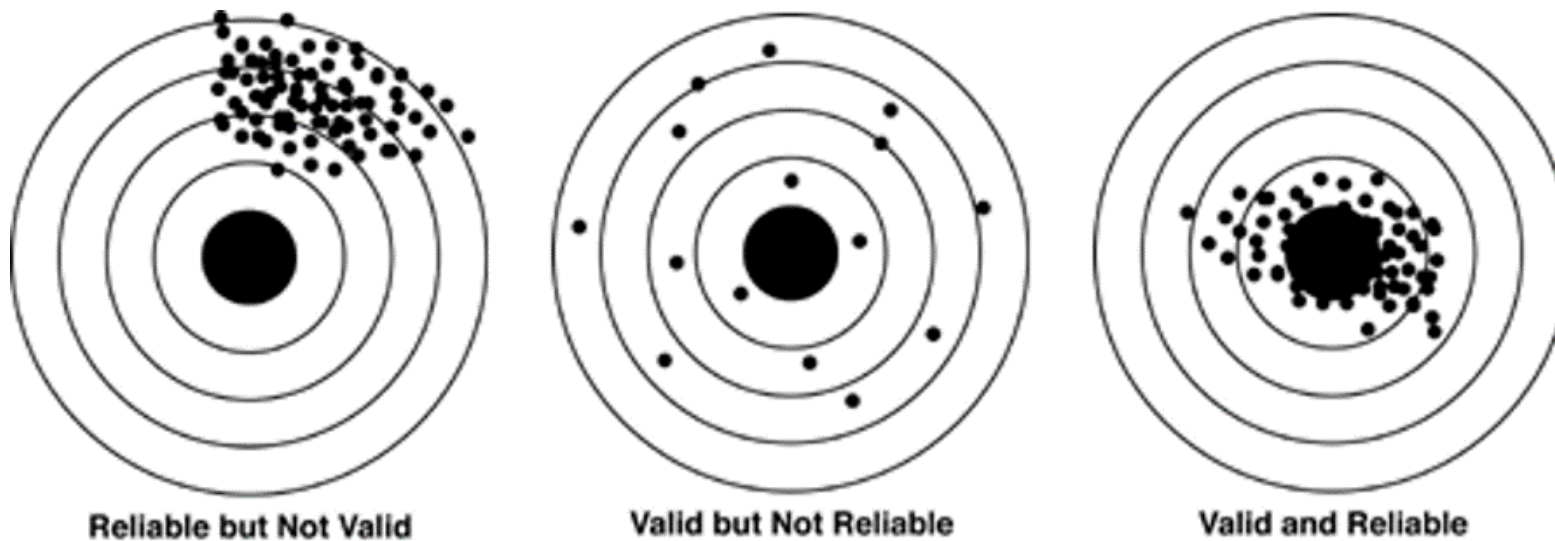
Challenges in Measurement

- + Measurement Error
 - + Random errors vs. Systematic errors
- + Bias in Measurement
 - + Observer bias, response bias, selection bias
- + Cultural and Linguistic Considerations
 - + Ensuring measures are valid across different cultures and languages
- + Ethical Considerations
 - + Privacy, consent, and responsible use of data



Wrapping Up...

The Importance of Good Measurement



Lingering Questions

- + How might the level of measurement affect the statistical analyses we can perform?
- + Can you think of examples where improper measurement could lead to incorrect conclusions?
- + How can we improve the reliability and validity of our measurements in real-world research?

